



STRATEGIC MARKET-ENTRY ASSESSMENT · DEFINITIVE EDITION

The Mycoprotein Opportunity

Sizing the prize in fungal biomass fermentation — and how to enter it, benchmarked against plant-based meat and plant protein

A full investor-grade thesis for a strategic entrant into commercial mycoprotein production. Covers market sizing, value chain and process economics, competitive structure, unit economics, capital flows, regulation across the US, Brazil and Europe, the GLP-1 and hybrid-meat demand vectors, Porter and SWOT analysis, scenarios, risks and a phased build/partner/buy playbook — 25 exhibits across 18 sections.

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Independent analysis · July 2026

Strictly private & confidential

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How to read this document

This is a decision document. Each section opens with the answer, then shows the evidence. The one-page thesis and scorecard are overleaf; the recommendation and phased playbook are in Section 17. Figures are triangulated from multiple third-party estimates and company disclosures and are presented as directional ranges — the alternative-protein data landscape is noisy, and where sources disagree we say so rather than manufacture false precision.

SECTION 00

Executive summary & investment scorecard

Mycoprotein has reached an inflection point. Historically a one-brand category (Quorn), it has diversified fast on the back of new fermentation strains and regulatory openings — but it is bifurcating sharply. Highly-capitalised **B2C** brands are failing under the weight of capex and customer-acquisition cost, while **B2B** ingredient platforms are unlocking scalable, cost-competitive growth. Our conclusion, reached independently and corroborated by a parallel analysis, is unambiguous: **enter — but as an industrial B2B ingredient / hybrid business anchored by cheap feedstock and secured offtake, never as another branded meat substitute.**

Bottom line up front

The consumer “fake-meat” retail thesis has broken; the fermentation-ingredient thesis has not. The two are routinely conflated — and that confusion is the opening. Lead with meat-analogue and **hybrid (50/50)** applications and high-value **GLP-1 / active-nutrition** formulations, co-locate a first plant of roughly **15–20k tonnes/year** with a carbohydrate feedstock source, and secure offtake before pouring concrete. Treat branded retail as an optional downstream, not the core bet.

Investment scorecard

Market attractiveness	FAVORABLE	Small but ~8–10% growth; huge headroom vs a \$2.3T meat market
Structural differentiation	STRONG	Beats meat and plant proteins on LCA; complete protein + fibre
Unit economics (at scale)	FAVORABLE	\$4–6/kg → ~\$2.5/kg beats beef/pork; not yet chicken
Capital intensity	HIGH RISK	\$80–150M per plant; 'valley of death' scale-up
Competitive intensity	MODERATE	B2B commodity-scale white space is still sparsely populated
Regulatory path	MIXED	US/Brazil/Singapore navigable; EU slow but credibility-conferring
Business-model risk	DECISIVE	B2C is a value trap; B2B/hybrid is the only viable route

VERDICT: GO — conditional on a B2B / hybrid model

Seven findings that frame the decision

- **A small niche with a structural edge.** Mycoprotein is ~\$0.76B (finished product) to ~\$1.4B (broad ingredient) today — a rounding error against a ~\$2.3 trillion meat market and the ~\$20B plant-protein pool — but it grows ~8–10% and, uniquely, beats *both* meat and plant proteins on emissions, land and water while delivering complete protein plus prebiotic fibre.
- **The monetisation has shifted.** Functional ingredients — supplements, feed, binders and the base for hybrid products — are now 70–80% of volume. Mycoprotein is no longer just a meat substitute; it is an ingredient platform.
- **Don't confuse the plant-based slump with the death of alt-protein.** US retail meat-analogue sales are down ~19% since 2023, yet global plant-based meat still grew ~8% and capital is rotating hard into fermentation (~75% of Q1 2026 alt-protein funding). The failure was a branded-CPG, retail-shelf failure — not a failure of the protein source.
- **Business model is the whole game.** Survivors are B2B suppliers with feedstock integration and offtake (ENOUGH, backed by Cargill). Casualties are asset-heavy branded players: Meati raised ~\$450M and sold for ~\$4M; Mycorena went bankrupt; Quorn books heavy losses on ~£187M of revenue.
- **Attractive economics, one binding constraint.** Biomass cost of production is ~\$4–6/kg falling toward ~\$2.5/kg at ~32k-tonne scale — below beef and pork, far below precision fermentation — because it harvests whole biomass with light purification. The constraint is capital intensity (\$80–150M/plant) and the scale-up “valley of death.” Feedstock cost and RNA-reduction yield are the biggest levers.
- **The highest-return niches are specialised.** The blockbuster demand vector is **GLP-1 companion nutrition** (nutrient-dense, high-protein, high-fibre foods for appetite-suppressed, muscle-losing users) at \$15–25/kg, alongside **hybrid 50/50 meat** (\$11.7B by 2034). These beat fighting for the \$2/kg commodity shelf.
- **Regulation is moat and map.** The US (GRAS) is fastest, Singapore a quick beachhead, Brazil now navigable (ANVISA RDC 839/2023), and the EU slow (2–6 years) but credibility-conferring with a 5-year data-exclusivity reward. Recent tightening — an FDA deficiency letter (Feb 2026) and new UK genome-sequencing/DIAAS rules (Mar 2026) — raises the bar and favours the well-advised.

SECTION 01

Context, scope and method

This assessment answers one question: **should a strategic investor enter commercial mycoprotein production over a 5–10-year horizon, and if so, how?** It is written for an investor comparing this opportunity against the wider plant-based and plant-protein markets, and it is deliberately even-handed about the category's headwinds.

Scope

- **Product.** Fungal biomass grown by fermentation for food — primarily mycoprotein in the strict sense (filamentous fungi such as *Fusarium venenatum*), plus adjacent fungal-biomass ingredients. Precision and gas fermentation appear only for contrast.
- **Applications.** Two demand vectors: mycoprotein as a high-value *ingredient* (supplements, GLP-1/active nutrition, binders) and as *biomass for meat-analogue and hybrid products*.
- **Geographies.** Brazil, the United States and Europe — the markets with the most relevant demand potential and the clearest (if uneven) regulatory routes.
- **Decision.** Whether to build, partner or buy; where to play; how to win; and how big a first plant should be.

Method at a glance

We triangulated published market estimates, company filings and disclosures, non-profit industry data (Good Food Institute), retail/consumption data (SPINS, Euromonitor), peer-reviewed life-cycle and techno-economic studies, and regulatory registers, and we cross-checked the result against an independent parallel analysis. Where third-party numbers diverge — and for market size they diverge by up to 5× — we present a range and a reasoned base case. All monetary figures are USD unless noted.

A note on data quality

Alternative-protein research is unusually noisy: definitions differ (does “mycoprotein” mean the ingredient, the finished product, or the whole meat-substitute shelf?), and vendors have incentives to publish large numbers. We flag the boundary for every figure and lean on primary disclosures — capacity, funding, filings — wherever possible. On sustainability in particular, headline multiples vs beef vary enormously with the baseline chosen; we report ranges, not single hero numbers.

SECTION 02

What mycoprotein is — and where it sits

Mycoprotein is protein-rich fungal biomass grown by fermentation and harvested whole. Quorn has grown *Fusarium venenatum* in steel tanks since the 1980s; a wave of newer entrants now grows other fungi by the same basic route. Understanding the fermentation taxonomy is essential, because the four approaches have **radically different economics** — and most of the investment noise (and most of the failures) sits outside biomass fermentation.

EXHIBIT 1

Four fermentation routes — mycoprotein is a biomass play

The economics differ by an order of magnitude across these rows.

Route	What it does	Representative players	Mycoprotein?
Traditional fermentation	Microbes transform a substrate (age/culture food)	Tempeh, koji, cultured dairy	Adjacent — not the focus
Biomass fermentation	Grow whole microbial cells fast; harvest the biomass itself	Quorn, ENOUGH/ABUNDA, Nature's Fynd, Meati, Fermotein, Rhiza, PEKILO	YES — mycoprotein lives here
Precision fermentation	Engineer microbes to secrete one specific molecule	Perfect Day (whey), Remilk (casein), Impossible (heme)	No — costlier, purification-heavy
Gas fermentation	Feed microbes CO ₂ / H ₂ / methane instead of sugar	Solar Foods (Solein), Calysta, Unibio	No — earlier-stage

Source: Independent analysis; company disclosures.

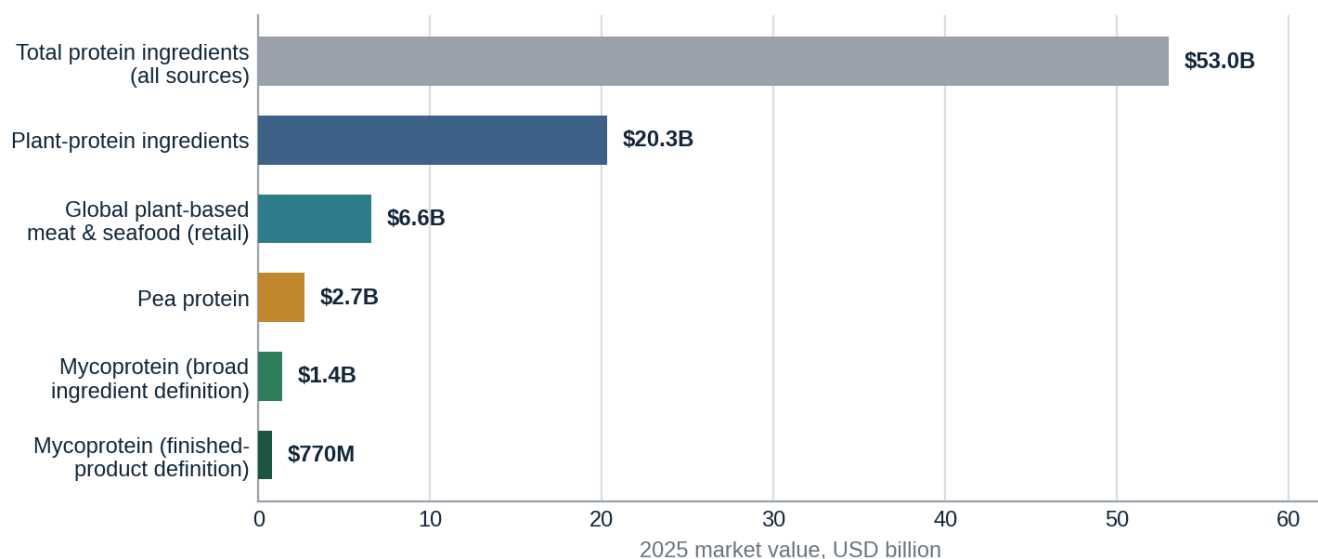
Why biomass fermentation is the sweet spot

Because biomass fermentation grows and harvests the *entire* cell rather than isolating and purifying a single molecule, its downstream processing is far lighter than precision fermentation's. That is the root of its cost advantage — and the reason mycoprotein can credibly chase parity with animal protein while precision fermentation still struggles against commodity ingredients. It also yields a whole-food ingredient (protein + fibre) rather than an isolate, which shapes both its nutritional positioning and its consumer story.

EXHIBIT 2

Mycoprotein is a sliver of the protein economy today

Where the category sits within the nested protein-market stack (2025).



Source: Independent analysis triangulating published estimates; GFI; Euromonitor. Boundaries differ by source; figures are directional.

The strategic read

Being small is the opportunity, not the problem. Mycoprotein is early enough that a single well-run 20k-tonne plant is a *material* share of global supply — unlike pea protein, where a new entrant is a price-taker against entrenched majors. The question is not “is the market big?” but “can we build a low-cost, reliable supply position before the category scales?”

SECTION 03

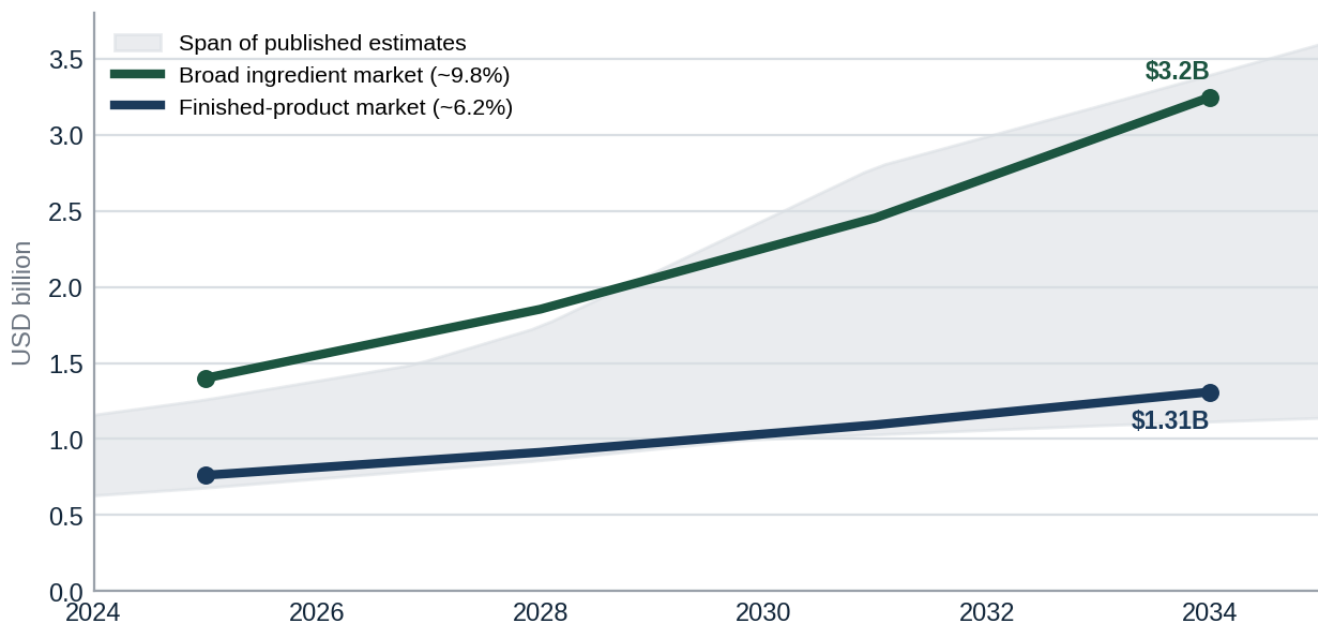
Market size and growth

The number you quote depends entirely on where you draw the boundary. The **finished-product** market is ~\$0.76B in 2025, growing ~6.2% to ~\$1.4B by the mid-2030s. The **broad ingredient** market — including supplements, feed and functional additives — is ~\$1.4B in 2025, growing ~9.8% to ~\$3.0B by 2034. Published estimates span an even wider cone. That spread is itself a finding: the trajectory hinges on whether the category escapes the branded-retail model dragging on the wider meat-analogue shelf.

EXHIBIT 3

Two defensible market definitions, one wide forecast cone

Global mycoprotein value; shaded band spans published third-party estimates.



Source: Independent analysis; band reflects five published estimate sets; two headline trajectories reflect finished-product vs broad-ingredient boundaries.

The monetisation has shifted toward functional ingredients

EXHIBIT 4

The monetisation has shifted toward functional ingredients

Approximate split of mycoprotein consumption by use.



Source: Independent analysis; industry estimates.

The takeaway for an entrant: **value is migrating from finished meat-analogues to ingredients** — supplements, hybrid bases and binders — which is exactly the higher-margin, lower-marketing terrain a B2B model should target.

Put in context against the comparison markets

- **vs plant-protein ingredients (~\$20B, ~8.5% CAGR).** Mycoprotein is tiny beside the pea/soy/wheat pool but growing at a similar-or-faster clip from a far smaller base — the classic profile of an emerging ingredient.
- **vs pea protein (~\$2.7B).** Even plant protein's current darling is only ~2–3.5× mycoprotein's size; mycoprotein's differentiation (fibre, complete EAAs, sustainability) is arguably sharper.
- **vs global plant-based meat & seafood (~\$6.6B retail).** Mycoprotein's most direct end-market; it grew ~8% globally in 2025 even as the US retail slice fell — demand is real but geographically uneven.
- **vs a ~\$2.3 trillion global meat market.** Penetration is microscopic, which is the bull case: the ceiling is enormous if mycoprotein competes as a mainstream protein input.

SECTION 04

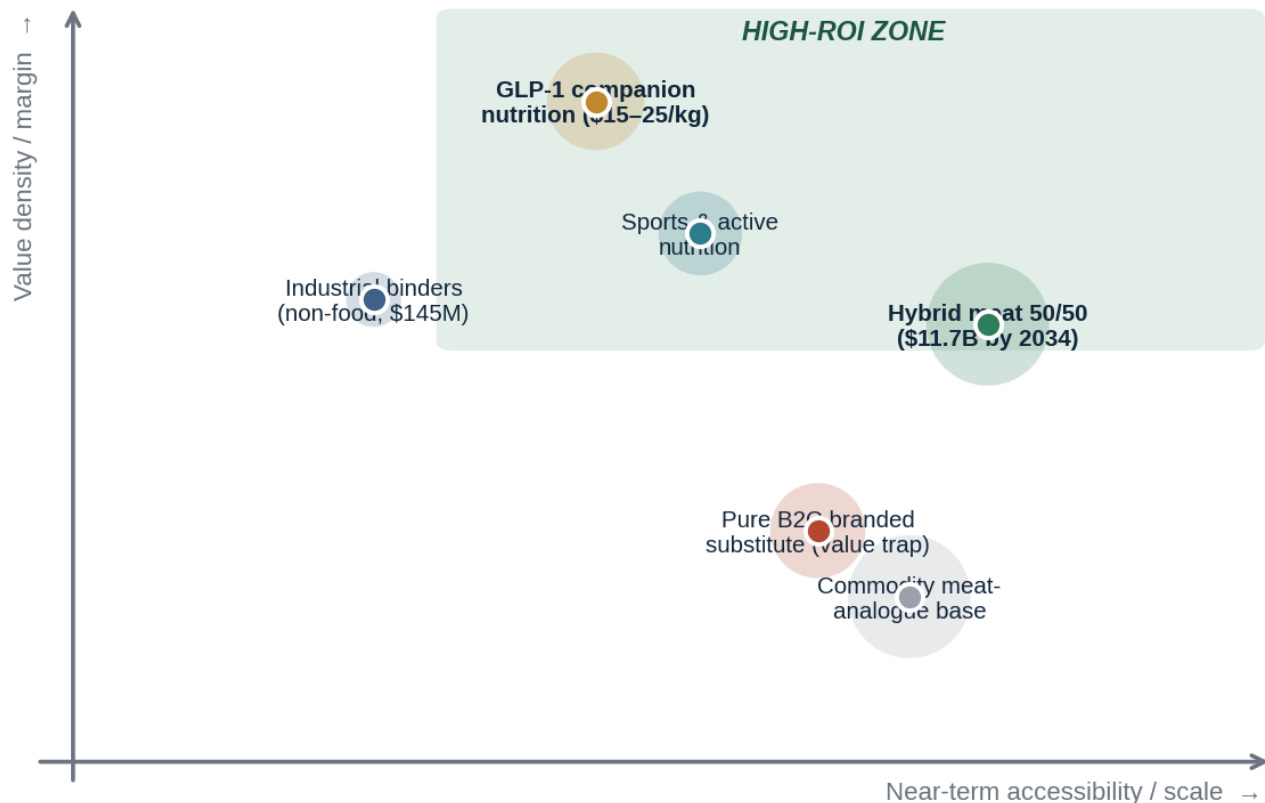
Where the value is moving — the niche map

The most viable paths to profit deliberately depart from the original dogma of replacing meat with 100% plant replicas. Three niches stand out, and two of them are where a disciplined entrant should concentrate.

EXHIBIT 5

Play the high-ROI niches, not the commodity shelf

Opportunity map by near-term accessibility and value density; bubble size $\approx$ market size.



Source: Independent analysis; market sizes from published estimates (hybrid meat $\sim$ \$11.7B by 2034; binders $\sim$ \$145M by 2036).

Hybrid meat — the pragmatic volume play

Blending mycoprotein with conventional meat (e.g. 50/50) sharply cuts carbon footprint, saturated fat and cost while preserving the familiar taste of animal fat and adding fibre. The hybrid-meat market is projected to reach **\$11.7B by 2034** ($\sim$ 10.4% CAGR). Crucially, it sidesteps consumer neophobia: it is sold to committed meat-eaters, not vegans. Quorn's own foodservice arm validated the model with its UltiMeat blended range in UK foodservice in 2025/26.

GLP-1 companion nutrition — the blockbuster vector

Why GLP-1 changes the demand picture

GLP-1 receptor agonists (Ozempic and successors) suppress appetite and can cause substantial muscle loss — in some patients **25–40% of weight lost is lean mass**. That creates urgent demand for nutrient-dense, high-protein, high-fibre foods in small portions — a description that fits mycoprotein almost exactly. These specialised formulations command **\$15–25/kg** versus ~\$2/kg for the commodity shelf, and the demand is price-insensitive. Tellingly, The Protein Brewery's recent round was led by Novo Holdings — the parent of Ozempic-maker Novo Nordisk.

Industrial binders — optionality outside food

Beyond food entirely, mycoprotein/mycelium is being explored as a sustainable binder in engineered wood, packaging and insulation — a replacement for toxic formaldehyde resins. The addressable market is small (~\$145M by 2036, ~17.9% CAGR) but sits *outside* food-safety regulation, offering a lower-friction revenue stream and a use for off-spec biomass.

SECTION 05

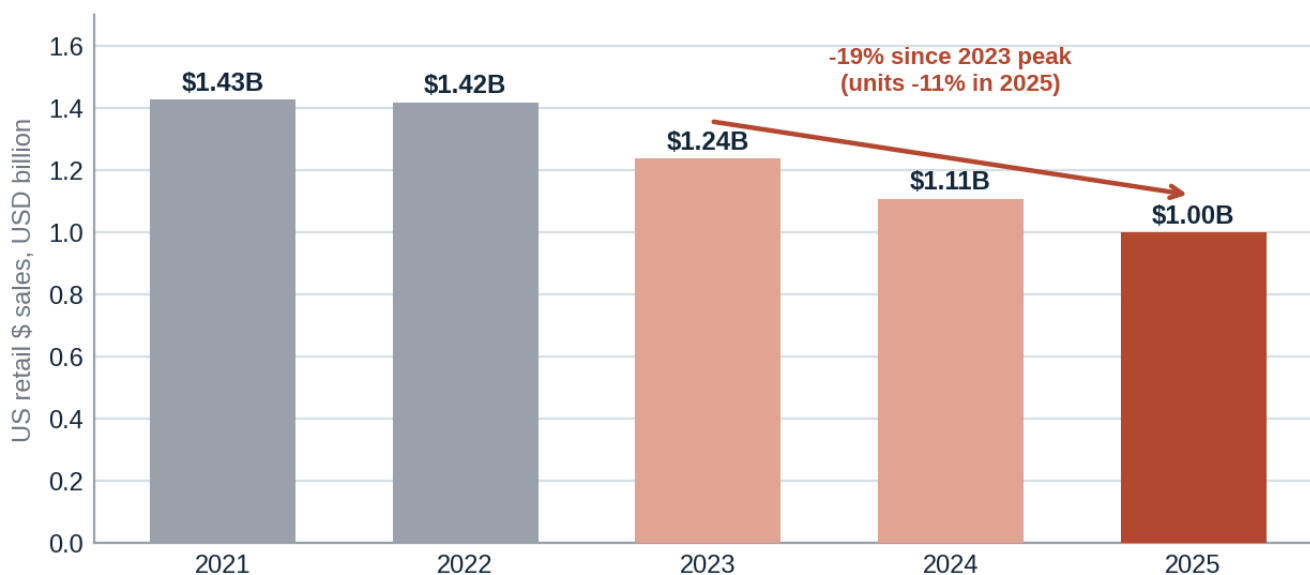
The plant-based reality check

Any entry thesis must confront the elephant in the room: branded plant-based meat has been in a multi-year slump in its most developed markets. This is the most important context for an investor — and the most misread. **The slump is a business-model and product problem, not proof that consumers have abandoned protein diversification.**

EXHIBIT 6

US retail alt-meat has fallen for three straight years

US retail dollar sales of plant-based meat & seafood (illustrative, triangulated).



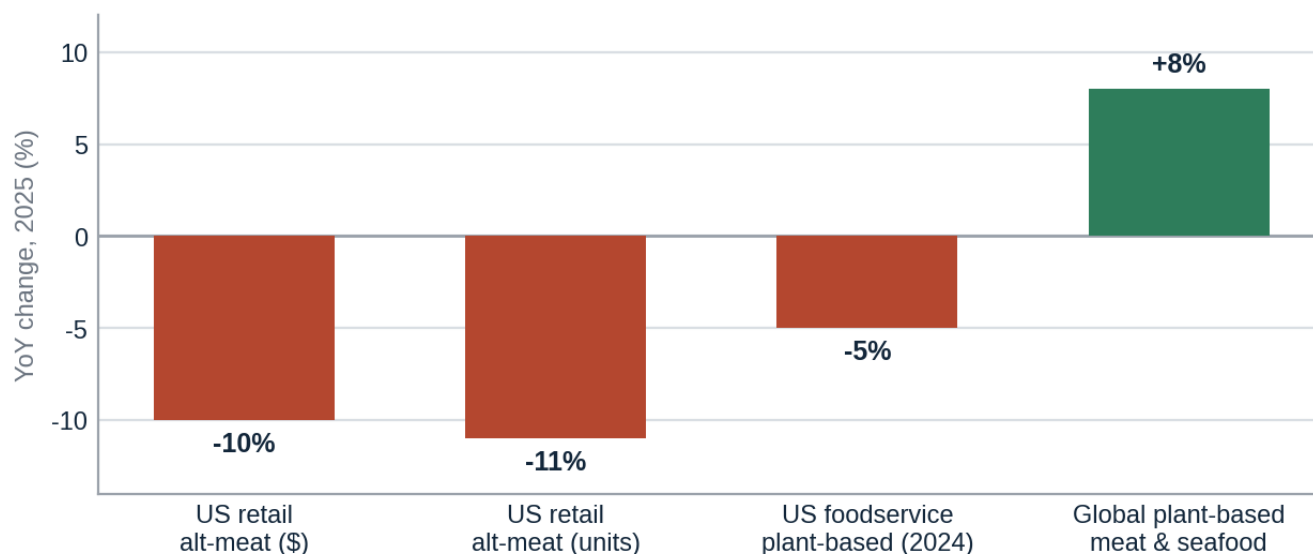
Source: Independent analysis anchored to GFI/SPINS data points (2025 ~\$1.0B, -10% YoY, units -11%; -19% since 2023). Annual path is illustrative.

The causes are well understood: premium pricing into an inflation-squeezed consumer, taste and texture that under-delivered, a growing “ultra-processed” perception, and over-distribution that outran genuine repeat demand. But the category is *repositioning*, not uniformly dying.

EXHIBIT 7

Not dying everywhere — repositioning

Divergent 2025 growth signals across channels and geographies.



Source: Independent analysis; GFI; SPINS; Euromonitor. US foodservice figure is 2024.

Plant protein still leads the substitute shelf — but its constraints are showing

EXHIBIT 8

Plant protein still leads the substitute shelf

Approximate share of the meat-substitute category.



Source: Independent analysis; industry estimates.

Why this helps a disciplined mycoprotein entrant

First, the reset has cleared weak competitors and cheapened assets and talent. Second, it validates the pivot the smart money is already making — away from own-brand retail toward **B2B ingredient supply, foodservice and hybrid products**, where mycoprotein's texture and nutrition actually win. An entrant who never builds a consumer brand sidesteps the exact trap that snared the category's pioneers.

SECTION 06

Demand drivers and the consumer landscape

Underneath the retail noise, the structural demand case is robust — but consumer-facing adoption faces real cultural friction that shapes *where* and *how* to sell.

The structural drivers

- **Sustainability as risk mitigation.** Animal agriculture drives ~a third of food-system GHG emissions and ~70% of freshwater use. As Scope-3 targets become binding for majors like Nestlé and Unilever, switching to mycoprotein becomes a supply-chain risk-management exercise, not a marketing choice.
- **Health megatrends.** The WHO classifies processed red meat as Group-1 carcinogenic; the “flexitarian” cohort — reducing meat without full vegetarianism — is now ~42% of adults in key markets, and is highly taste- and texture-sensitive, favouring mycoprotein’s natural fibrousness.
- **The GLP-1 wave.** As covered in Section 4, appetite suppression plus muscle loss is generating unprecedented demand for nutrient-dense high-protein foods.

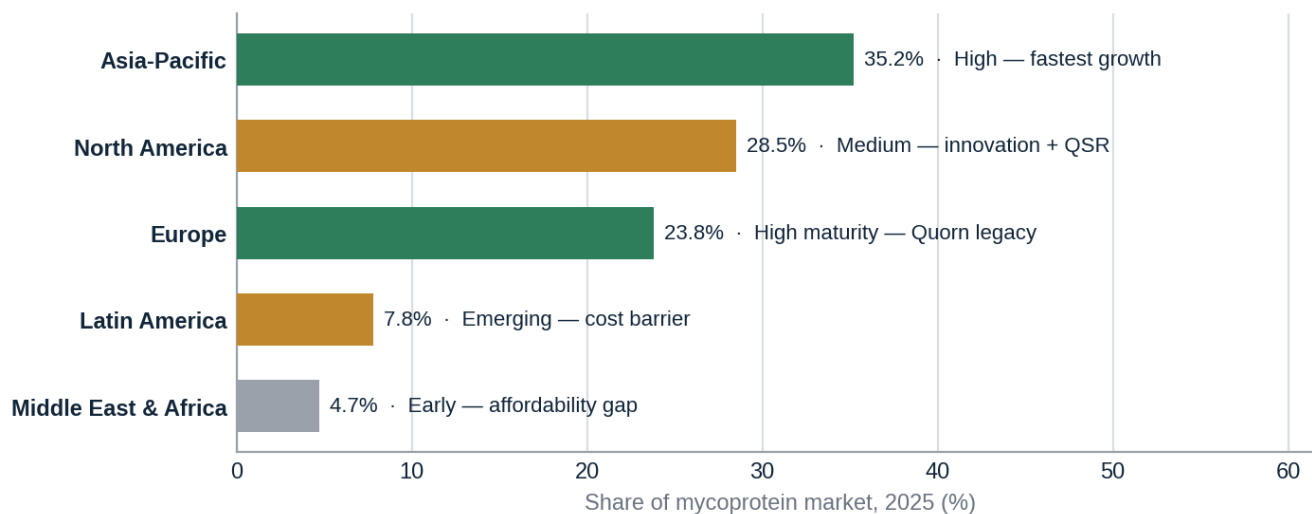
The consumer barriers — and the regional map

The core friction is semantic: “fungus,” “biomass” and “microbe” trigger visceral food neophobia in many Western consumers (only ~16% report knowingly eating “fungus meat”), pushing brands toward opaque naming like “fermented protein.” Acceptance varies sharply by region — which should shape market sequencing.

EXHIBIT 9

Acceptance and market share vary sharply by region

Mycoprotein market share and adoption posture by region (2025).



Source: Independent analysis; published regional estimates.

SECTION 07

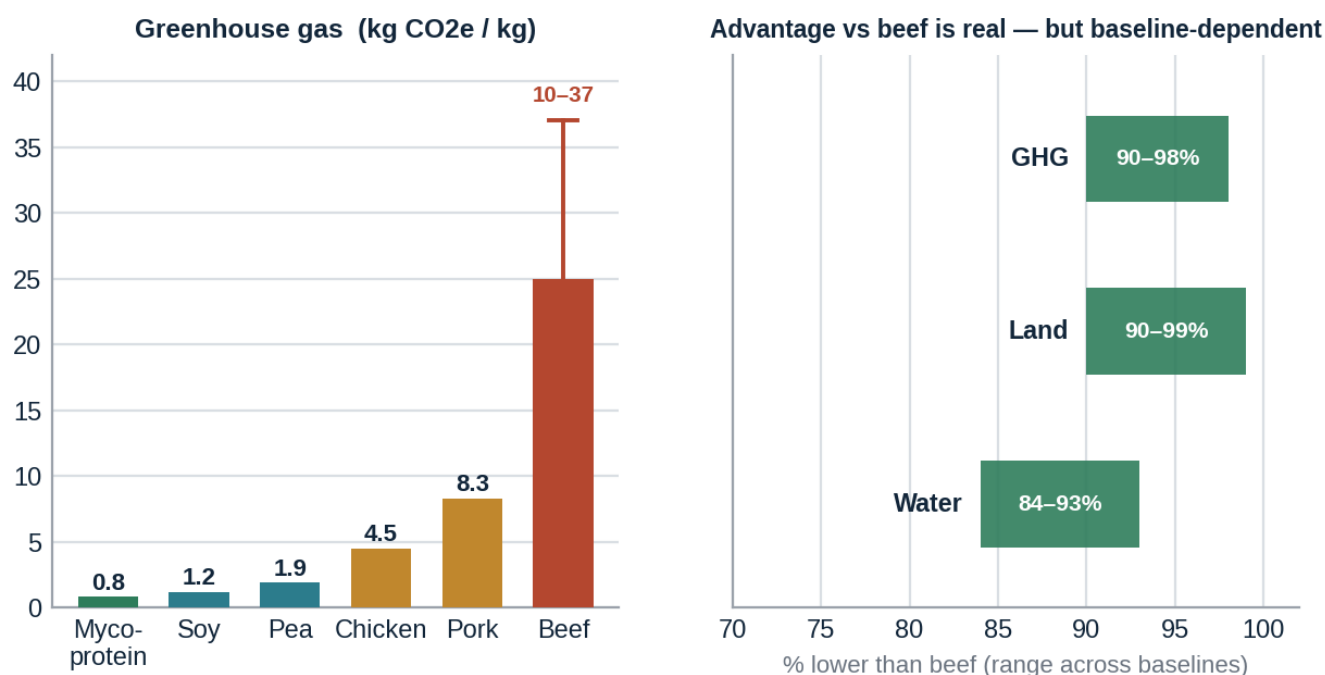
Why mycoprotein is structurally different

Mycoprotein is not just “another plant-based option.” On the two axes investors should care about most — **sustainability** and **nutrition** — it holds a defensible, evidence-backed edge over both animal *and* plant proteins. That edge is the durable reason the category outlives the current downturn.

EXHIBIT 10

Lower footprint than meat — and than plant proteins

Cradle-to-gate impacts, and the (baseline-dependent) advantage over beef.



Source: Peer-reviewed LCAs and systematic reviews; producer footprint reports. Beef GHG spans 10–37; advantage vs beef varies with the baseline chosen.

The honest sustainability headline

Mycoprotein emits roughly **0.7–0.8 kg CO₂e/kg** — lower than soy and pea protein, and far lower than any meat. Its advantage over beef is genuinely large but *baseline-dependent*: peer-reviewed reviews put it near **90% lower**, while producer LCAs against grass-fed beef reach **~98% (up to ~55×)**. Land and water advantages are similarly wide (roughly 6–13.5× less water, 10–80× less land). The safe, defensible framing for a data room: **dramatically lower on all three, exact multiple depends on the beef system.**

Nutrition — strong, but be honest about protein density

Mycoprotein is a complete protein (all nine essential amino acids) with a **PDCAAS of ~0.996** — equivalent to milk — plus prebiotic fibre, no cholesterol and low saturated fat, and it avoids the antinutrients (phytates, lectins) and amino-acid gaps (e.g. pea's methionine, wheat's lysine) that dog some plant proteins. Its natural hyphal structure recreates meat's bite without the harsh high-moisture extrusion that globular plant proteins require. But on *raw protein density* the isolates win — whey/soy/pea deliver ~80–90 g/100g versus mycoprotein's ~45–50 g. The supplement play must therefore be a differentiated *protein-plus-fibre* proposition, not a head-to-head on grams per dollar.

EXHIBIT 11

Whole-food strength; isolate-grade density is not the pitch

Attribute	Mycoprotein	Pea isolate	Soy isolate	Whey
Protein (per 100 g, dry)	~45–50 g	~80 g	~85–90 g	~80 g
Dietary fibre	High (prebiotic)	Low	Low	None
Complete amino acids	Yes (all 9)	Near (low Met)	Yes	Yes
PDCAAS	~0.996	~0.89	~0.91	~1.00
Antinutrients	None notable	Phytates	Phytates	None
Processing to fibre	Natural hyphae	HME extrusion	HME extrusion	n/a (animal)
Footprint vs peers	Lowest	Low	Low	High

Source: Independent analysis of typical published compositions; values vary by product and preparation.

SECTION 08

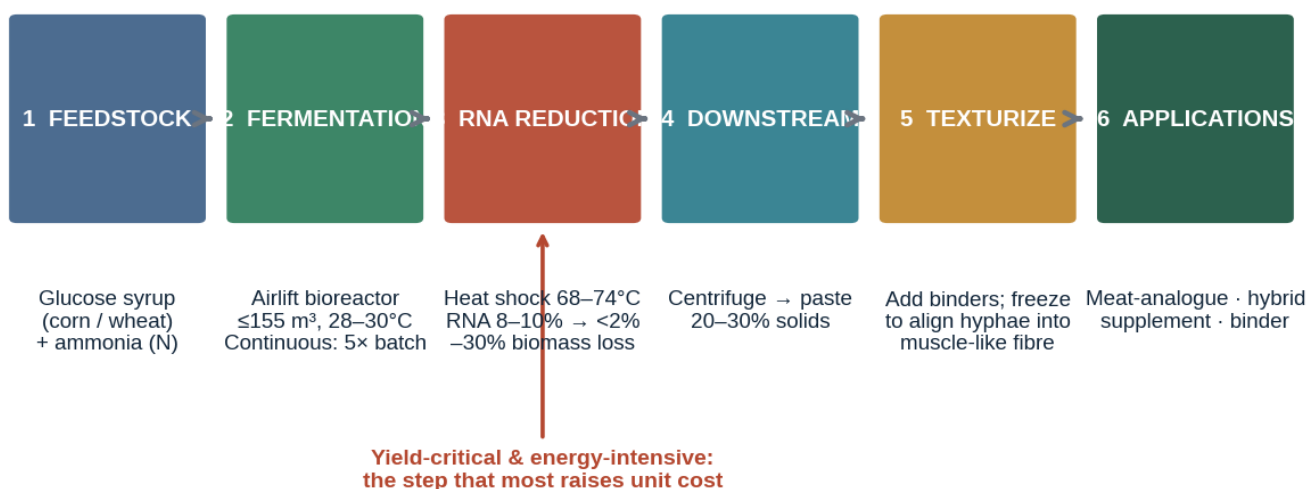
The technology and value chain

Mycoprotein's economics are decided upstream, in a handful of process steps. Understanding them is not academic — two of these steps (RNA reduction and strain stability) are where unit cost and operational risk actually live, and they explain why the category is capital- and know-how-intensive.

EXHIBIT 12

From sugar to muscle-like fibre — and where the cost hides

Simplified biomass-fermentation value chain with key parameters.



Source: Independent analysis; process parameters from published mycoprotein literature and patents.

The two steps that decide the economics

- **RNA reduction is the yield killer.** Fast-growing fungal biomass contains 8–10% RNA; consumed untreated it converts to uric acid (gout, kidney stones), so the broth must be heat-shocked (68–74°C, 30–45 min) to activate RNases and bring RNA below the WHO's 2% limit. This step is energy-hungry and destroys up to **30% of the final solid biomass** — a direct hit to unit economics. (The released nucleotides do have umami value and can be sold as flavour enhancers.)
- **Strain mutation forces costly stoppages.** In continuous culture, a branched *Fusarium* mutant tends to emerge and out-compete the parent after ~100 generations (~6 weeks), degrading the linear-filament structure that makes the meat-like texture — forcing a shutdown, sterilisation and re-inoculation cycle. Managing this is core operational know-how.
- **Continuous fermentation is a requirement, not a nicety.** It delivers up to **5×** the productivity of batch operation and is essential to approach parity with conventional meat; airlift bioreactors up to 155 m³ run at 28–30°C under tight pH and oxygen control.
- **Feedstock and formulation.** Glucose syrup (corn/wheat) plus ammonia is the base; the centrifuged paste (20–30% solids) is bound (egg albumin for vegetarian; potato protein, gluten or alginate for vegan) and frozen to align hyphae into fibre. The circular-economy frontier — using agro-industrial sidestreams as feedstock (e.g. Enifer's PEKILO) — both cuts opex and strengthens the ESG story.

SECTION 09

The competitive landscape

The field splits into three tiers by production maturity and strategy. The most important competitive fact for an entrant: the **B2B, commodity-scale corner is still sparsely populated** — ENOUGH is effectively alone at scale, with Enifer scaling behind it.

EXHIBIT 13

The mycoprotein field: who's who

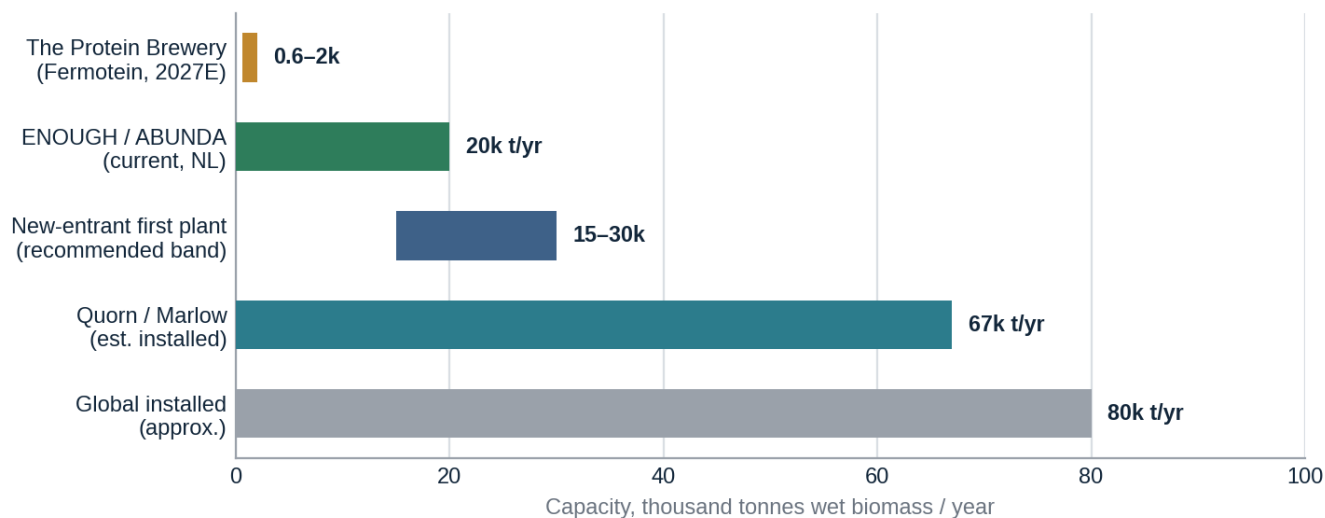
Player	Organism & process	Go-to-market	Capacity	Ownership & funding	Status / notes
Quorn / Marlow (Monde Nissin)	<i>F. venenatum</i> ; biomass	Branded CPG + nascent B2B	~67k (est.)	Acq. £550M (2015); heavy losses	Leader ~24–31% global; £187M rev (2024, -9%); QuornPro + hybrids
ENOUGH / ABUNDA	<i>F. venenatum</i> ; biomass	B2B ingredient	20k (site ≥60k)	~€95M+; Cargill stake + offtake	~13–15% share; co-located with Cargill; Unilever, M&S;
Nature's Fynd	<i>F. flavolapis</i> (Fy); surface ferment.	Branded + ingredient	n/d	\$500M+ (Gates, Bezos, Gen. IM)	US GRAS 2021; premium; capital-heavy
Meati (Emergy)	Mushroom mycelium; submerged	Branded whole-cut	"mega" plant (CO)	~\$450M → sold ~\$4M	Cautionary tale: covenant sweep → distress sale → revival
The Protein Brewery	<i>R. pusillus</i> (Fermotein)	B2B ingredient	0.6k → 2k (2027–29)	~\$60M; Novo Holdings	First novel mycelium approved in EU (2026); active-nutrition focus
Enifer	PEKILO; circular fermentation	B2B ingredient	Scaling 2026	~€49M; €33M plant	Agro-sidestream feedstock; Kantvik (FI); pet food + human
Also / distressed	Various fungi	Mixed	Early / n/a	Sub-scale	Mycorena (bankrupt → Naplasol); Better Meat (Rhiza, GRAS 2024); MyForest

Source: Company disclosures, filings and press; independent analysis. "n/d" = not disclosed. Capacities are wet-biomass estimates, t/yr.

EXHIBIT 14

Sizing intuition: where a first plant should land

Installed / target capacity benchmarks, thousand tonnes wet biomass per year.



Source: Company disclosures; independent analysis. Global installed ~80k is an approximation.

EXHIBIT 15**The attractive white space is B2B at commodity scale**

Strategic map by go-to-market and positioning; bubble size ≈ scale/funding.



Source: Independent analysis.

How to win

Play where ENOUGH and Enifer play — **B2B ingredient supply at commodity scale** — and out-execute on the two things that decide this market: **delivered cost** (feedstock integration) and **application performance** (win hybrid and poultry-style uses where mycoprotein's fibre and texture are unmatched). Brand is not the battleground; reliable, low-cost, spec-consistent supply is.

SECTION 10

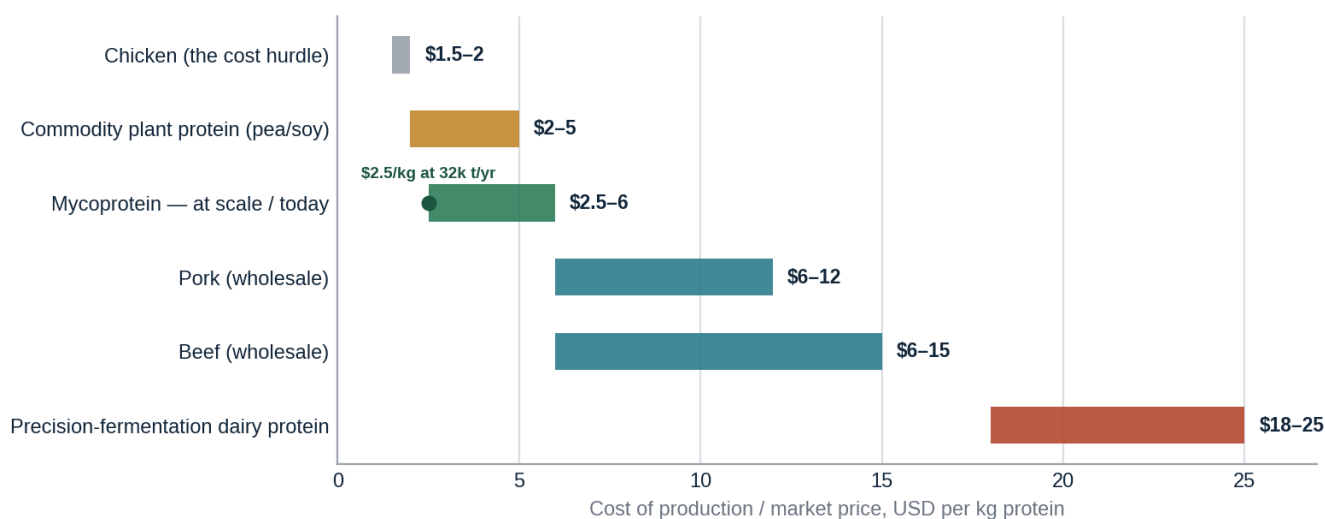
Value chain and unit economics

Here is the good news the consumer struggles obscure: **biomass fermentation has credible unit economics**. Published techno-economic models put cost of production around **\$4–6/kg**, falling to ~\$2.5/kg at ~32k-tonne commercial scale — below wholesale beef and pork, and far below purified precision-fermentation proteins. The honest caveat: it does not yet beat **chicken** (~\$1.50–2.00/kg), which is the real cost benchmark.

EXHIBIT 16

Biomass beats beef and pork; chicken is the real hurdle

Cost of production / market price, USD per kg protein.



Source: GFI techno-economic insights; published TEA models; commodity price ranges.

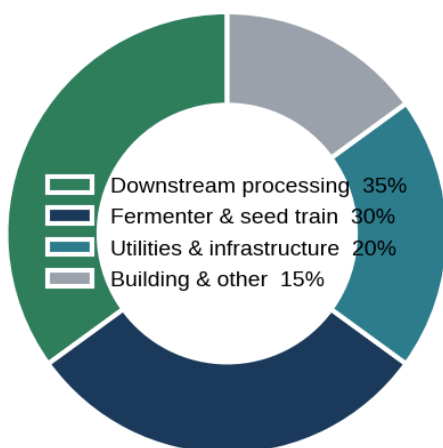
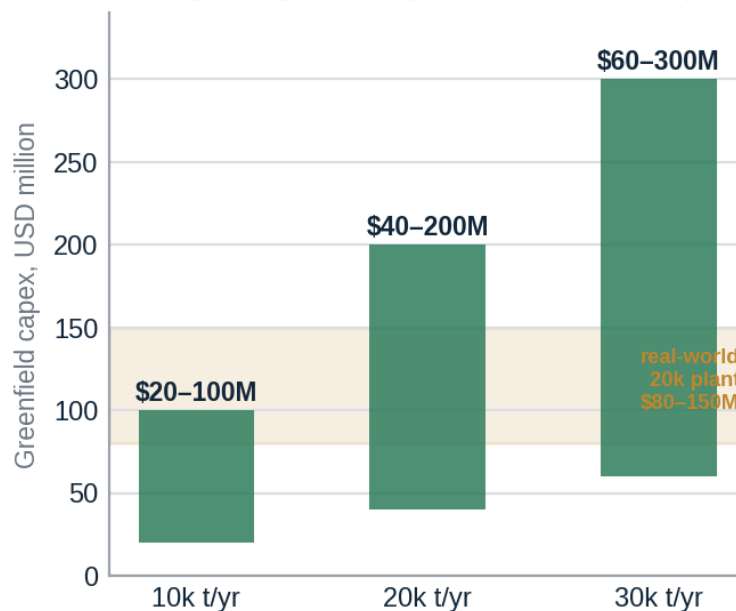
The binding constraint is capital, not cost-per-kilo

The reason mycoprotein has not scaled faster is not that a kilogram is expensive to make — it is that **getting to first commercial kilograms is capital-intensive and operationally hard**. A greenfield plant for 10–20k t/yr runs **\$80–120M (often \$150M+ post-Covid)**; the fermenter itself is only ~30% of that, with downstream processing another ~35%. More than 60% of biomanufacturing ventures that succeed in the lab never reach commercial scale — the “valley of death” that took Meati despite \$450M of funding.

EXHIBIT 17

Capex structure and the plant-size trade-off

Left: typical capex composition. Right: greenfield capex by plant size.

Fermentation plant capex mix**Capex vs plant size (\$2–10k / annual tonne)**

Source: Published bioprocess TEA rules of thumb; independent analysis.

The one lever that matters most: feedstock

Feedstock (fermentable carbohydrate) is ~a third of cash cost, and the models are clear that viability needs a carbon source below ~\$0.40/kg and high conversion yield. This is **exactly** why ENOUGH co-located with a Cargill starch plant — taking cheap sugars in and sending waste streams to an adjacent bioethanol unit in a zero-waste loop, sharing even waste process heat. Any entrant with privileged access to low-cost carbohydrate and industrial utilities starts with a structural margin advantage a pure-play startup cannot replicate.

SECTION 11

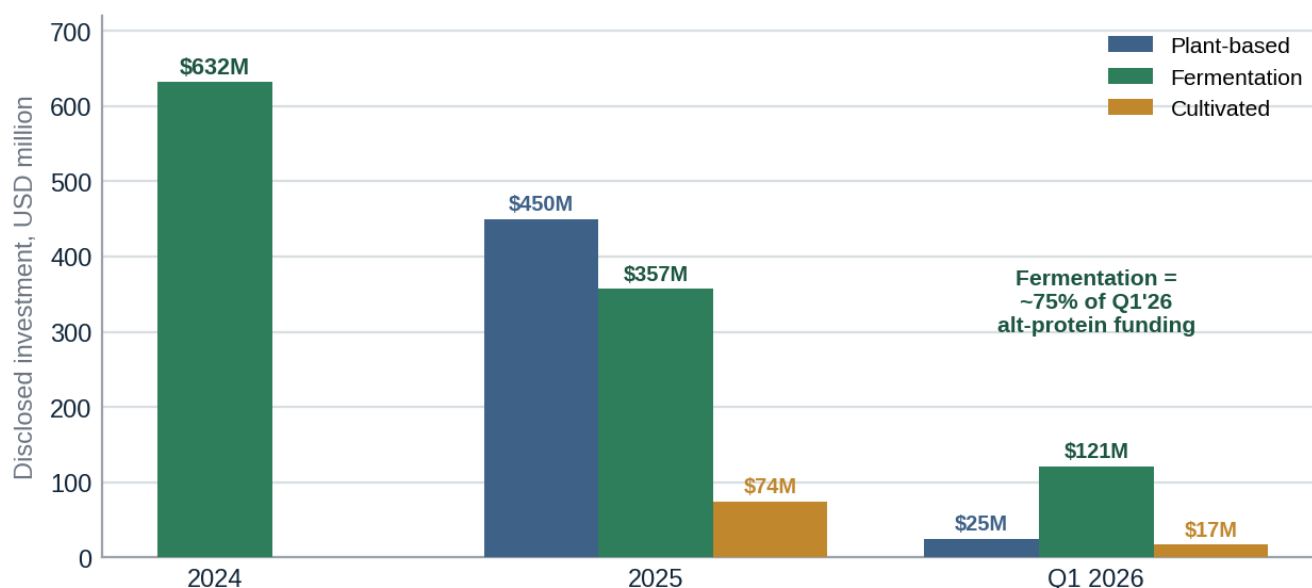
The capital picture

Money is voting for fermentation. Total alternative-protein funding was ~\$881M in 2025 (cumulative >\$19B since 2016). Within that, **fermentation now takes the lion's share of new capital — ~75% of a modest Q1 2026 total** — even as absolute fermentation funding cooled from its 2024 peak. The signal is a flight to the sub-sector with the clearest path to real production.

EXHIBIT 18

Capital is rotating into fermentation — on tougher terms

Disclosed alternative-protein investment by sub-sector.



Source: GFI analysis of Net Zero Insights data. 2024 shown for fermentation only.

Two things are true at once. The **direction** is favourable — fermentation is where sophisticated capital concentrates, and four of the five largest 2024 alt-protein rounds were fermentation companies. But the **terms** have hardened: the market has moved “from promise to proof,” and high-profile failures (Meati in biomass, Motif in precision, Arkeon in gas) mean investors now underwrite scale-up risk, offtake and unit economics — not just science.

Implication for a strategic entrant

A well-capitalised strategic investor is *advantaged* here. The companies dying are those dependent on fickle venture rounds and covenant-laden debt to fund capex. An entrant that can (a) self-fund a staged plant, (b) bring feedstock and offtake, and (c) take a 7–10-year view can acquire distressed assets and talent cheaply and build the low-cost position the venture-funded field cannot.

SECTION 12

Regulation as moat and map

Regulatory approval is a real barrier to entry — which makes it a moat once cleared. Routes differ sharply across the three target geographies, and sequencing them well can save years. The headline: **start where approval is fastest and cheapest, use it to build evidence, then tackle the slow-but-credible markets.**

EXHIBIT 19

A patchwork — sequence entry from easy to hard

Relative favourability of the regulatory pathway by geography.

	Pathway speed	Low cost / burden	Mycoprotein precedent	Market attractiveness
United States	Favorable	Favorable	Favorable	Favorable
Singapore	Favorable	Favorable	Favorable	Moderate
Brazil	Moderate	Moderate	Moderate	Favorable
Canada	Moderate	Moderate	Favorable	Moderate
European Union	Challenging	Challenging	Favorable	Favorable

Source: Regulatory registers and filings (FDA GRAS; EU Novel Food 2015/2283; Singapore SFA; Health Canada; ANVISA RDC 839/2023); independent analysis.

- **United States — fastest, but tightening.** GRAS has cleared Quorn, Nature's Fynd (2021), ENOUGH and Rhiza (2024). But the FDA has hardened its stance: it issued a **deficiency letter in February 2026** that forced The Protein Brewery to withdraw its Fermotein GRAS notice pending animal-study and antinutrient clarification. Self-affirmed GRAS is under political pressure — file early and build a full dossier.
- **Singapore — fast beachhead.** A proactive novel-foods regime (multiple mycoprotein approvals) makes it an efficient first-approval and credibility market despite its size.
- **Brazil — newly navigable.** ANVISA's RDC 839/2023 established a defined novel-foods pathway at end-2023 — a material improvement for a large, protein-hungry market with domestic feedstock advantages.
- **European Union — slow but credible, with a reward.** The Novel Food Regulation is the world's most stringent (Fermotein took ~6 years), but Quorn is grandfathered (pre-1997), Fermotein has now opened the door, and the EU grants the authorised producer **5-year data exclusivity** — a formidable barrier against fast followers.
- **United Kingdom — rising evidentiary bar.** FSA guidance (March 2026) now makes whole-genome sequencing and DIAAS digestibility testing effectively mandatory — more cost and time to budget for.

SECTION 13

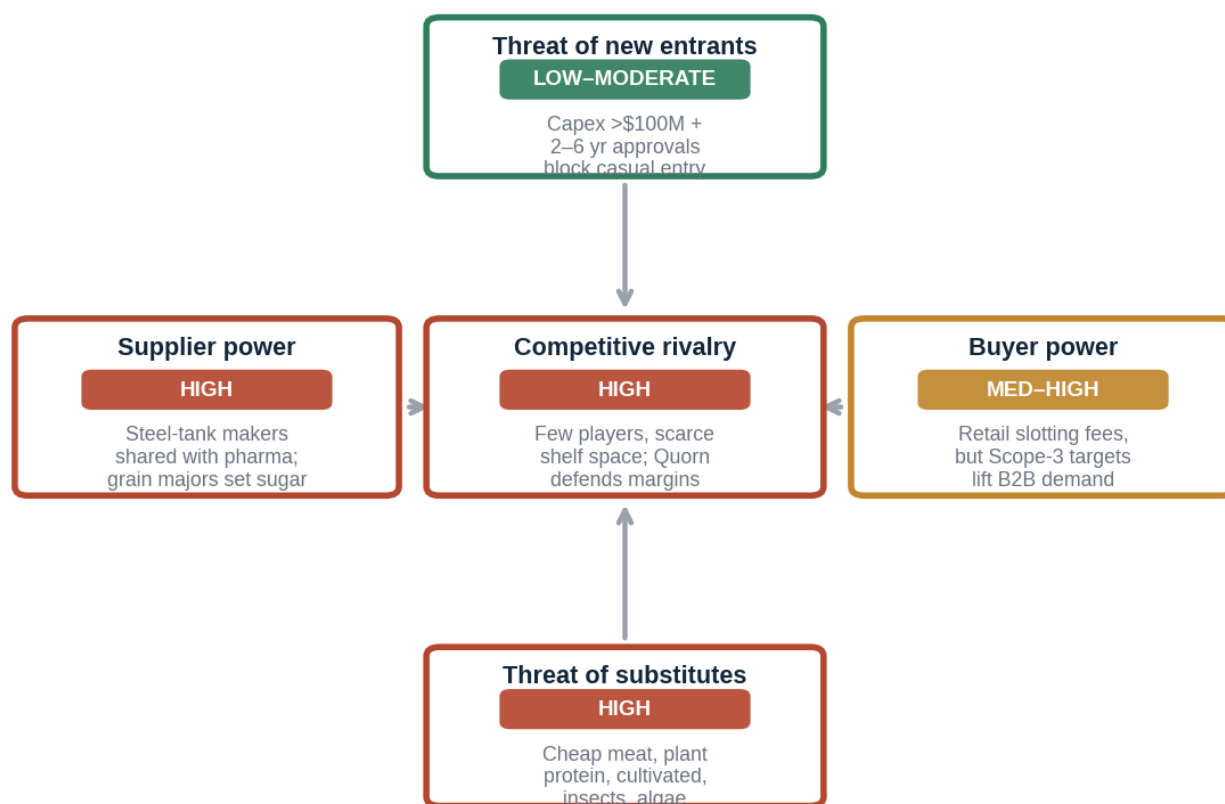
Industry attractiveness — Porter's Five Forces

Structurally, the mycoprotein industry is **moderately attractive and defensible for a well-capitalised incumbent**, precisely because the same forces that make entry hard (capital, regulation, supplier power) also protect those already inside.

EXHIBIT 20

The five forces favour a scaled, integrated entrant

Competitive-force intensities and their drivers.



Source: Independent analysis (Porter framework).

- **New entrants (low–moderate):** \$100M+ bioreactor capex and 2–6-year global approval labyrinths block trivial entry — good news for a serious entrant, bad news for casual ones.
- **Supplier power (high):** sterile steel-tank capacity is shared with Big Pharma (premium pricing), and glucose supply sits with grain-processing giants — an argument for feedstock integration.
- **Buyer power (medium–high):** retail slotting fees squeeze margins, but Fortune-500 Scope-3 urgency gives B2B suppliers leverage to lock in stable annual offtake.
- **Substitutes (high):** cheap subsidised meat, plant protein, and emerging cultivated/insect/algae platforms all compete for the same capital and attention.
- **Rivalry (high):** few players, scarce shelf space, and Quorn's gravitational pull force price discipline — reinforcing that cost leadership, not brand, is the winning axis.

SECTION 14

SWOT synthesis

The integrated view: mycoprotein's **strengths and opportunities are structural and durable** (sustainability, nutrition, GLP-1, hybrid, feedstock integration), while its **weaknesses and threats are largely executional and macro** (capex, RNA-yield drag, regulatory timing, perception) — which is exactly the risk profile a disciplined industrial operator is best placed to manage.

EXHIBIT 21

SWOT — durable upside, executional risk

STRENGTHS

- Organic fibrous texture without harsh chemical extrusion
- Best-in-class LCA — radical cuts in carbon, water, land
- Complete protein + prebiotic fibre; PDCAAS ~0.996
- Closed-system production, insulated from climate & zoonotic shocks

OPPORTUNITIES

- Capitalise on the GLP-1 & active-ageing megatrend (\$15–25/kg)
- Hybrid 50/50 formulations for QSR & foodservice
- Feedstock co-location + upcycling (ENOUGH–Cargill model)
- B2B diversification into industrial binders & materials

WEAKNESSES

- Overwhelming capex to reach genuine scale
- Unit-economics drag from RNA reduction (~30% biomass loss)
- Neutral base flavour needs formulation to mimic meat
- Strain-mutation cycles force costly production stoppages

THREATS

- Unpredictable FDA/EFSA timelines stalling cash flow
- Structural capital crunch felling under-funded rivals
- 'Ultra-processed' perception & consumer fatigue
- Continued decline in retail chicken prices

Source: Independent analysis.

SECTION 15

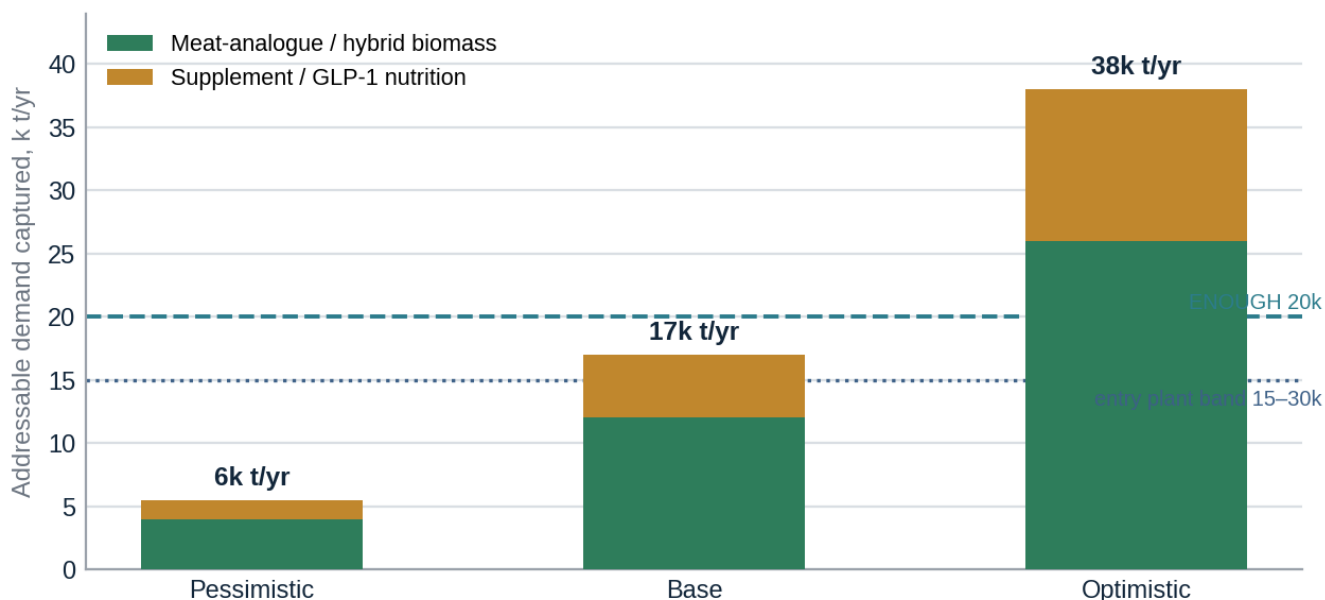
Demand scenarios and entry sizing

Translating context into a plant is the crux. We frame demand along the two application vectors and size a first plant to the **share of addressable demand a disciplined entrant can credibly capture**. The figures are an *illustrative sizing framework*, anchored to real capacity benchmarks; calibrate the exact numbers to an internal demand model and binding offtake.

EXHIBIT 22

Base case points to a ~15–20k t/yr first plant

Illustrative addressable demand captured by a single entrant, by scenario and vector.



Source: Independent sizing framework; capacity benchmarks from company disclosures.

- **Base case (most likely, ~9–10% CAGR):** steady B2B and flexitarian demand; mycoprotein establishes itself as a premium functional ingredient; the broad market reaches ~\$3.0B by 2034. A ~15–20k t/yr first line is the right size — large enough to be cost-competitive and a meaningful share of global supply, small enough to stage capex and de-risk scale-up.
- **Optimistic:** faster FDA/EFSA clearance and lower rates unlock large capex into >100k-litre plants; costs fall below chicken parity; hybrids go mainstream at major retailers; market approaches ~\$4B.
- **Pessimistic:** prolonged supply-chain friction and a venture-funding freeze deepen the capex “valley of death,” felling more innovators (as in 2024–25); “artificial” perception persists.
- **Sequence to the regulatory map:** base the first plant to serve US and Brazilian demand (fast/navigable approval, feedstock access); treat Europe and the GLP-1 premium niche as Phase-3 expansions once approval and scale economics are in hand.

SECTION 16

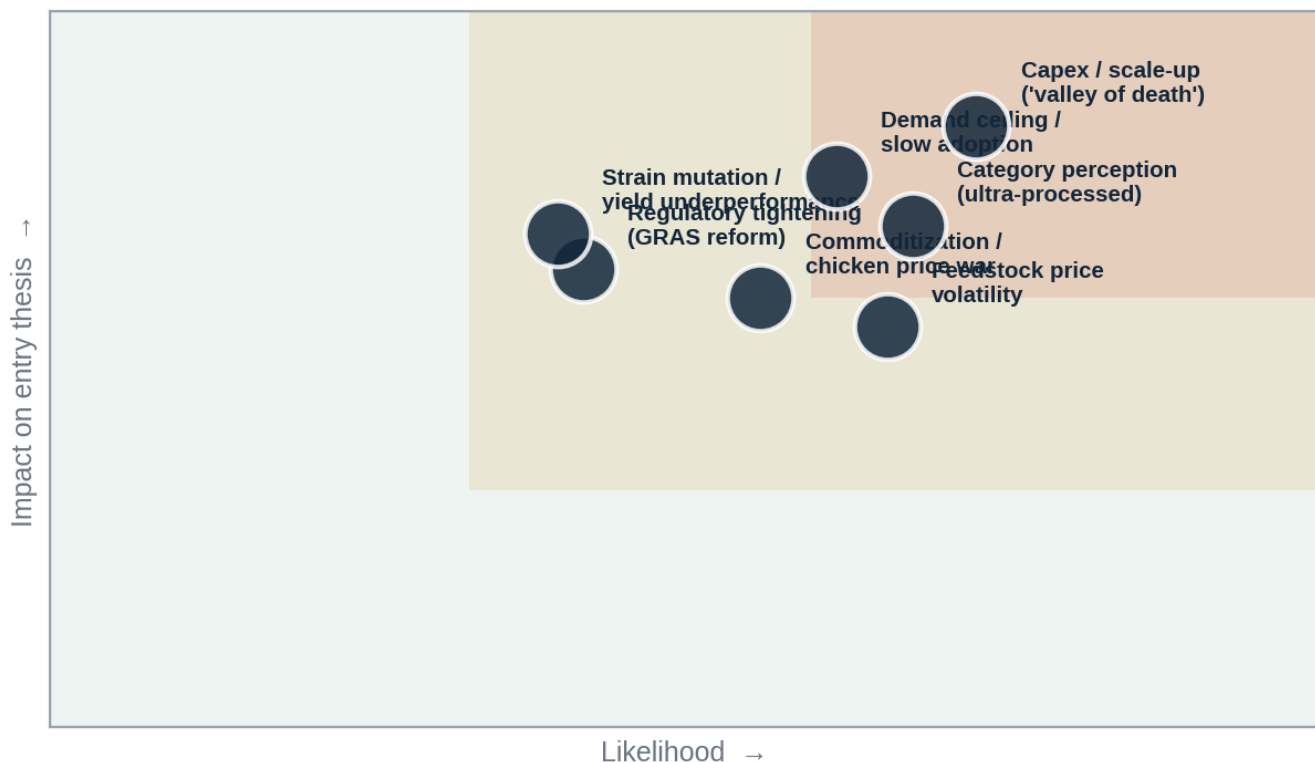
Risks and sensitivities

The dominant risks cluster in execution and demand — exactly where an experienced industrial operator with feedstock and offtake has the most control. The risks an entrant controls *least* (perception, regulatory reform) are real but second-order to getting the plant built and sold-out.

EXHIBIT 23

Execution and demand dominate the risk map

Key risks by likelihood and impact on the entry thesis.



Source: Independent analysis.

Mitigations

- **Capex / scale-up:** stage capacity; license a proven process rather than inventing one; partner with an operator who has already scaled a fermentation line.
- **Demand ceiling:** de-risk with binding offtake before construction; anchor to a large customer's reformulation programme; keep the plant flexible across applications and the GLP-1 niche.
- **Feedstock & RNA-yield:** secure long-term carbohydrate supply and co-locate; invest R&D; in low-RNA, thermotolerant strains to attack the single biggest cost/yield drag.
- **Strain mutation:** treat continuous-culture stability as core IP; budget planned stop/sterilise/re-inoculate cycles into opex.
- **Perception:** lean on the whole-food, minimal-processing, high-fibre story; favour hybrid and foodservice channels less exposed to front-of-pack scrutiny.
- **Regulatory:** file early under current rules; diversify approvals across geographies so no single regime is a chokepoint.

SECTION 17

Strategic options and the recommendation

Three business models are on the table — and the data is decisive about which works in this cycle.

- **Vertically-integrated B2C (unviable now):** build the plant, build the brand, and educate the consumer through marketing — the “triangle of death” that Meati’s burn-rate collapse illustrates. Only Quorn survives here, subsidised by a parent.
- **B2B ingredient supplier (recommended):** ENOUGH and The Protein Brewery are the model — decouple from marketing, optimise volumetric yield, and use large offtake agreements as financial collateral for debt. ENOUGH’s Cargill co-location dramatically cut logistics and feedstock capex.
- **Hybrid-meat formulator (tactical expansion):** blend 30–50% mycelium into conventional meat to cut emissions and add fibre without alienating meat-eaters; QuornPro’s UltiMeat foodservice deal validated it in 2025/26.

EXHIBIT 24

Build vs partner vs buy — partner-then-build wins

Route	Speed	Capex exposure	Control / IP	Key risk	Fit for entrant
Build (cold greenfield)	Slow	Highest	Full	Scale-up & demand risk borne alone	Weak on its own
Partner / offtake / JV	Fast	Shared / staged	Shared	Partner dependency	RECOMMENDED core
Buy (distressed assets)	Fast	Low (cents on \$)	High	Integration & tech fit	Strong accelerant

Source: Independent analysis.

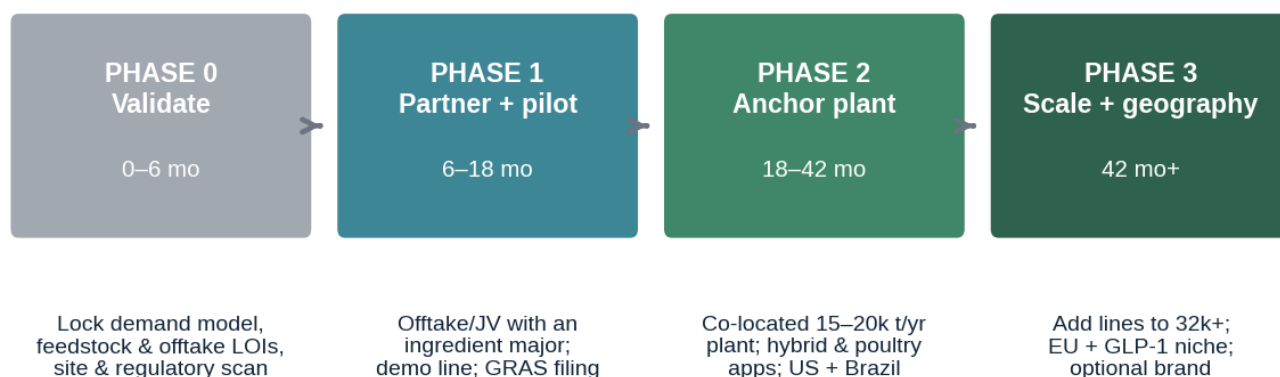
Where to play / how to win — in one line each

- **Where to play:** B2B mycoprotein ingredient supply at commodity scale plus the high-margin GLP-1/active-nutrition niche — not branded retail meat substitutes.
- **How to win:** lowest delivered cost (feedstock co-location + staged capex + low-RNA strains) and best application performance (hybrid and poultry-style first).
- **Where to build:** co-located with carbohydrate feedstock, serving US + Brazil demand first; Europe (and its 5-year exclusivity) as a later, credibility-conferring expansion.
- **How big:** a first line of ~15–20k t/yr, expandable toward 32k+, sized to binding offtake.

A phased playbook

EXHIBIT 25

A four-phase entry playbook — validate, partner, anchor, scale



Source: Independent analysis.

The recommendation, restated

Proceed — conditionally, and as an ingredient / hybrid business. Commit to Phase 0–1 now: lock the demand model, secure feedstock and offtake letters of intent, choose a co-located site, and begin a US GRAS filing with a full dossier. Gate the Phase-2 capital commitment on binding offtake covering the majority of first-line volume. Redirect what a B2C strategy would spend on brand and retail into industrial capex, low-RNA strain R&D, and the GLP-1/active-nutrition formulations where returns are highest. Do *not* enter as a branded meat-substitute company — that is the one version of this bet the evidence says to avoid.

SECTION A

Methodology, assumptions and sources

Approach

Figures were triangulated across independent categories of evidence, cross-checked against a parallel independent analysis, and reconciled to reasoned base cases. Market-size ranges reflect the spread of published third-party estimates; where those diverge materially, we show the range and a central case. Capacity, funding and financial figures rely on primary company disclosures and filings where available. Sizing figures in Section 15 are an explicit illustrative framework to be calibrated against the investor's own demand model and offtake commitments.

Key assumptions

- Mycoprotein 2025 market ~\$0.76B (finished product) to ~\$1.4B (broad ingredient); base-case growth ~9–10% (published range 5–17%).
- Biomass cost of production ~\$4–6/kg protein, falling to ~\$2.5/kg at ~32k t/yr; greenfield capex ~\$80–150M for a 10–20k-tonne plant (~\$2,000–10,000 per annual tonne).
- First-plant sizing band 15–20k t/yr, benchmarked to ENOUGH (20k) and Quorn (~67k est.).
- LCA values are cradle-to-gate and vary by system boundary; beef GHG spans a wide 10–37 kg CO₂e/kg, and the advantage-vs-beef multiple varies accordingly.
- RNA-reduction biomass loss ~30%; strain-mutation cycle ~100 generations (~6 weeks) in continuous culture; continuous vs batch productivity up to ~5×.

Principal sources

The Good Food Institute (State-of-the-Industry and investment analyses, drawing on Net Zero Insights); SPINS and Euromonitor retail/consumption data; company disclosures, annual accounts and press (Monde Nissin/Marlow Foods, ENOUGH, Nature's Fynd, Emergy/Meati, The Protein Brewery, Enifer, The Better Meat Co, Cargill); peer-reviewed life-cycle assessments, techno-economic models and bioprocess-engineering references (including work on mycoprotein RNA reduction and continuous fermentation); market-research syntheses (Precedence, FMI, Fact.MR, DataM, Persistence, IndexBox and others); and regulatory registers (US FDA GRAS notices incl. GRN 1268; EU Novel Food Regulation 2015/2283 and EFSA opinions; Singapore Food Agency; Health Canada; Brazil ANVISA RDC 839/2023; UK FSA guidance). Specific figures are directional and current to mid-2026.

Important notice

This document was prepared as an independent strategic analysis for discussion purposes. It is not investment, legal, tax, accounting or medical/nutritional advice, and it does not constitute an offer or recommendation to buy or sell any security or asset. Market figures are estimates triangulated from third-party sources believed to be reliable but not independently verified; actual results may differ materially. Any investment decision should rely on independent due diligence and professional advice. For health, physiological or GLP-1-therapy questions, consult a qualified medical professional.